

Module 08 Lab - The Bias-Variance

Tradeoff **Objective:** To understand and visualize the concepts of **overfitting**, **underfitting**, and the **bias-variance tradeoff**, which are central to building models that generalize well to new data. **In this lab, you will train models of varying complexity and plot their performance to see these concepts in action.**

Part 1: Understanding the Concepts* **Underfitting (High Bias):** The model is **too simple** and fails to capture the underlying patterns in the data. It performs poorly on both the training data and the test data. It has high bias because it makes strong, incorrect assumptions about the data.* **Overfitting (High Variance):** The model is **too complex** and learns the training data too well, including the noise and random fluctuations. It performs exceptionally well on the training data but poorly on the test data because it has memorized the training set instead of learning the general pattern. It has high variance because its performance changes drastically with different training data.* **The Goal:** Find a model that is "just right"—complex enough to capture the true pattern but not so complex that it memorizes the noise. This is the **bias-variance tradeoff**.

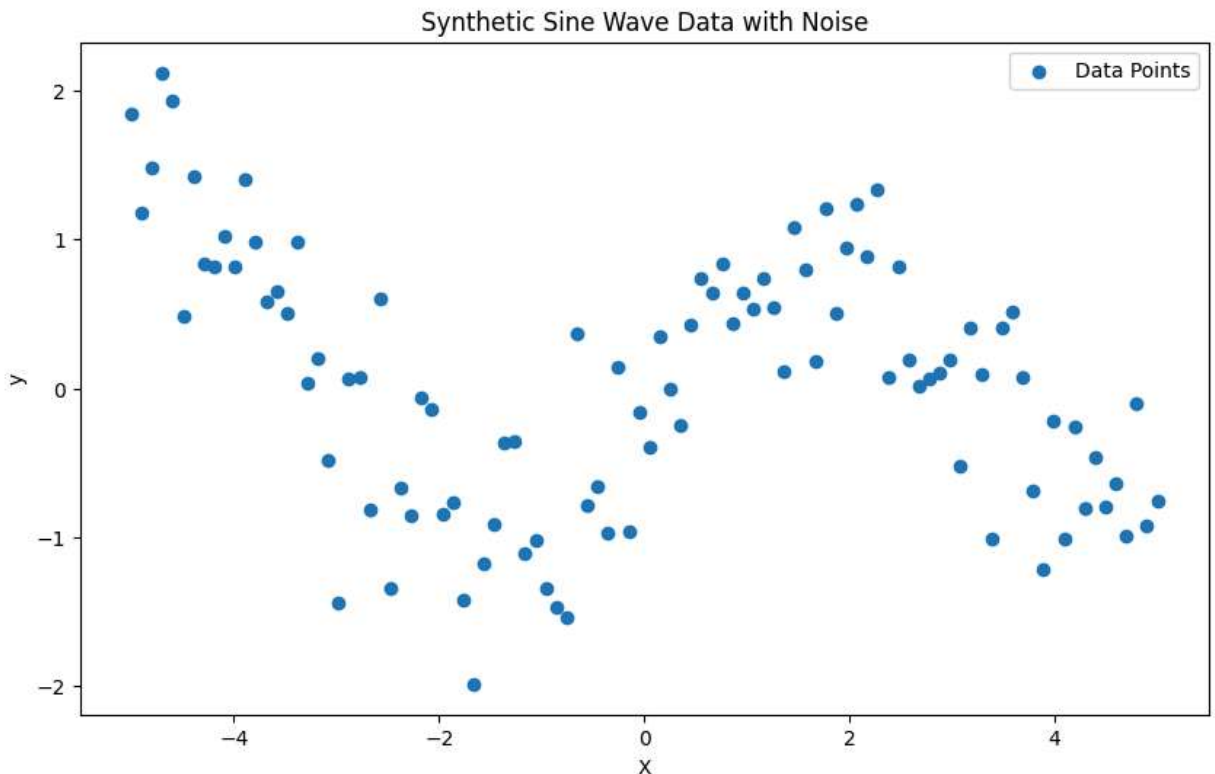
Part 2: Setup To visualize this, we will create a synthetic (fake) dataset with a known pattern and some noise.

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import learning_curve

# Generate some synthetic data
```

```
np.random.seed(0)
X = np.linspace(-5, 5, 100)
y = np.sin(X) + np.random.normal(0, 0.5, 100)
X = X[:, np.newaxis]

# Plot the data to see the underlying sine wave pattern
plt.figure(figsize=(10, 6))
plt.scatter(X, y, label='Data Points')
plt.title('Synthetic Sine Wave Data with Noise')
plt.xlabel('X')
plt.ylabel('y')
plt.legend()
plt.show()
```



Part 3: Modeling with Different Complexities We will use

Polynomial Regression to control model complexity. A

polynomial of degree 1 is a simple straight line (underfitting).

- ✓ A polynomial of degree 15 is a very complex, wiggly line (overfitting). **Your Task:** Train and visualize three models with different degrees (1, 4, and 15) to see underfitting, a good fit, and overfitting.

Task 1: Train and Plot Models **Your Task:** For each degree (1, 4, 15),

- ✓ create a polynomial regression model, train it, and plot its predictions against the original data.

```
# Task 1: Train and plot polynomial regression models
plt.figure(figsize=(14, 8))
plt.scatter(X, y, label='Data Points')

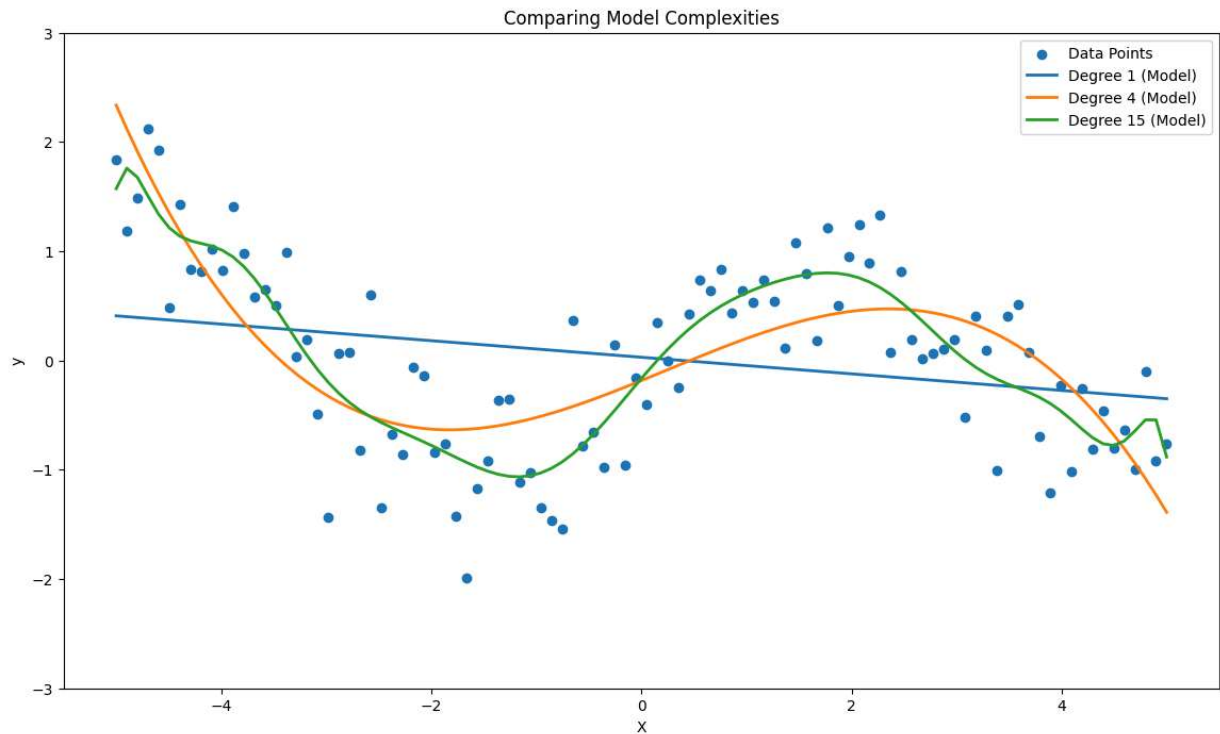
degrees = [1, 4, 15]
for degree in degrees:
    # 1. Create a pipeline that adds polynomial features and fits linear regres
    model = make_pipeline(PolynomialFeatures(degree), LinearRegression())

    # 2. Train the model
    model.fit(X, y)

    # 3. Make predictions
    y_pred = model.predict(X)

    # 4. Plot the model's predictions
    plt.plot(X, y_pred, linewidth=2, label=f'Degree {degree} (Model)')

plt.xlabel('X')
plt.ylabel('y')
plt.legend()
plt.title('Comparing Model Complexities')
plt.ylim(-3, 3)
plt.show()
```



Part 4: Learning Curves **Concept:** A **learning curve** is a powerful tool to diagnose bias and variance. It plots the model's performance (e.g., accuracy or error) on both the **training set** and the **validation set** as a function of the

- ✓ number of training samples.* **High Bias (Underfitting):** Both the training score and validation score will be low and will plateau quickly. The model is too simple to learn from more data.* **High Variance (Overfitting):** There will be a large gap between the high training score and the low validation score. The model memorized the training data but can't generalize.* **Just Right:** The training and validation scores will converge to a high value.

Task 2: Plot Learning Curves **Your Task:** Use the `learning_curve`

- ✓ function from scikit-learn to plot the learning curves for the underfit (degree 1) and overfit (degree 15) models.

```
def plot_learning_curve(estimator, title, X, y, ylim=None, cv=None,
                        n_jobs=None, train_sizes=np.linspace(.1, 1.0, 5)):
    plt.figure(figsize=(10, 6))
    plt.title(title)
    if ylim is not None:
        plt.ylim(*ylim)
    plt.xlabel("Training examples")
    plt.ylabel("R² score")

    train_sizes, train_scores, test_scores = learning_curve(
        estimator,
        X,
        y,
        cv=cv,
        n_jobs=n_jobs,
        train_sizes=train_sizes,
        shuffle=True,
        random_state=0
    )

    train_scores_mean = np.mean(train_scores, axis=1)
    train_scores_std = np.std(train_scores, axis=1)
    test_scores_mean = np.mean(test_scores, axis=1)
    test_scores_std = np.std(test_scores, axis=1)

    plt.grid()
    plt.fill_between(
        train_sizes,
        train_scores_mean - train_scores_std,
        train_scores_mean + train_scores_std,
        alpha=0.1,
        color="r"
    )
    plt.fill_between(
        train_sizes,
```

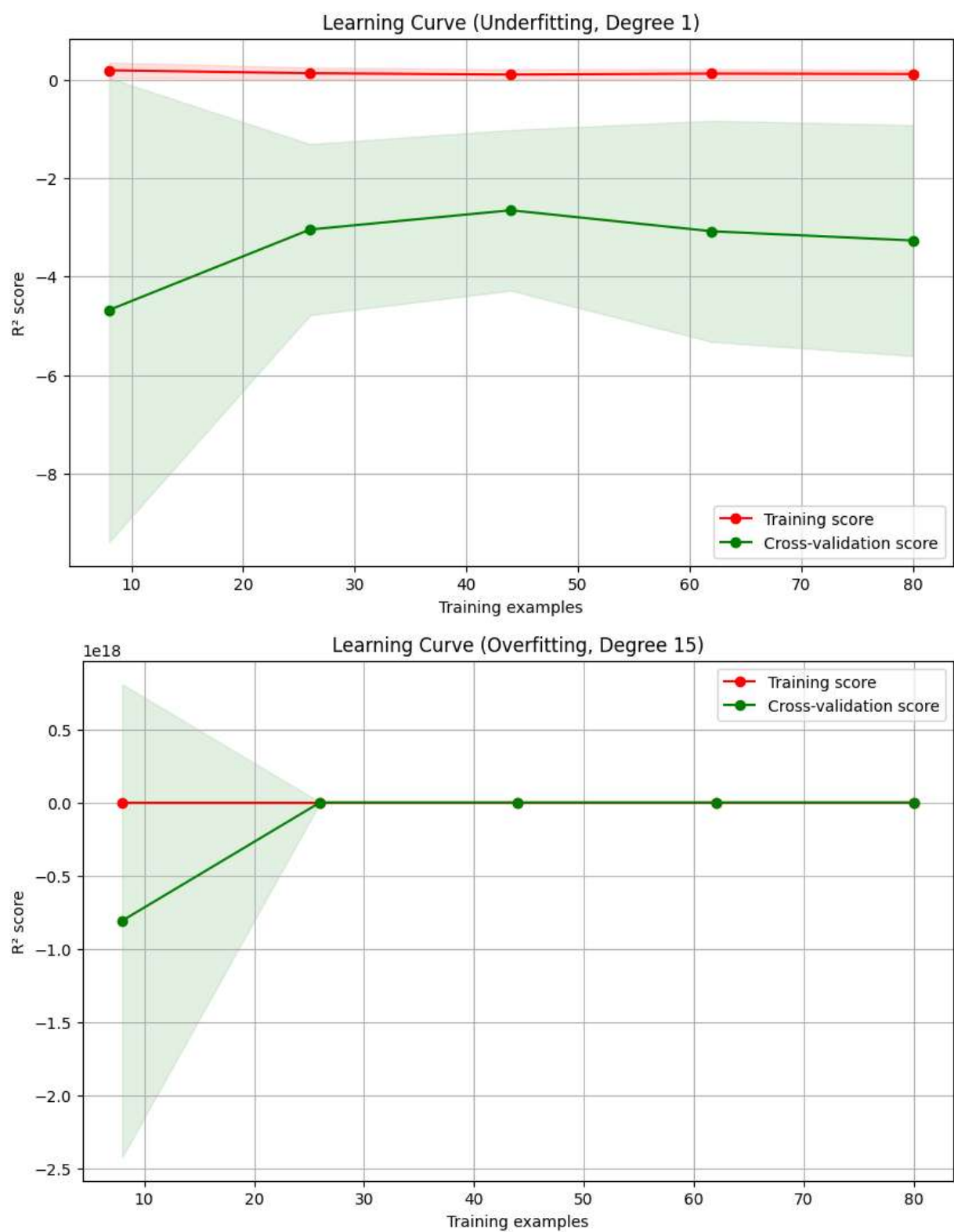
```
        test_scores_mean - test_scores_std,
        test_scores_mean + test_scores_std,
        alpha=0.1,
        color="g"
    )
    plt.plot(
        train_sizes,
        train_scores_mean,
        'o-',
        color="r",
        label="Training score"
    )
    plt.plot(
        train_sizes,
        test_scores_mean,
        'o-',
        color="g",
        label="Cross-validation score"
    )

    plt.legend(loc="best")
    return plt

# 1. Create the underfit and overfit models
underfit_model = make_pipeline(PolynomialFeatures(1), LinearRegression())
overfit_model = make_pipeline(PolynomialFeatures(15), LinearRegression())

# 2. Plot the learning curve for the underfit model
plot_learning_curve(
    underfit_model,
    "Learning Curve (Underfitting, Degree 1)",
    X,
    y,
    cv=5
)
plt.show()

# 3. Plot the learning curve for the overfit model
plot_learning_curve(
    overfit_model,
    "Learning Curve (Overfitting, Degree 15)",
    X,
    y,
    cv=5
)
plt.show()
```



Knowledge Check

1. In the first plot of the three models, which model is underfitting, overfitting, and a good fit? Explain your reasoning.

The degree 1 model is underfitting because it is only a straight line and cannot follow the curved sine-wave pattern. The degree 15 model is overfitting because its line bends sharply to follow small changes and noise in the training data. The degree 4 model is the best fit of the three because it follows the main pattern without reacting too strongly to every noisy data point.

2. **Looking at the learning curve for the underfitting model, what do you observe about the training and cross-validation scores? What does this tell you?**

The training and cross-validation scores are both relatively low and stay close together as more training examples are added. This tells me the degree 1 model has high bias. It is too simple to learn the real pattern, so adding more data does not improve it very much.

3. **Looking at the learning curve for the overfitting model, what do you observe about the gap between the training and cross-validation scores? What does this tell you?**

The degree 15 model has a much higher training score than cross-validation score, especially with smaller training sets. The large gap shows high variance. The model learns the training data very closely, including noise, but does not generalize as well to new data. As more training examples are added, the gap may become